AgriScan

AI-Based Banana Leaf Disease Detection

Submitted in partial fulfillment of the requirements
of the course **Innovative Product Development-III**

Year 3, Sem V Computer Engineering

By

Vishvesh Shrikant	60004230173
Vedang Kakade	60004230259
Vaishvee Gandhi	60004230081

Guide:

Dr. Kiran Bhowmick
Associate Head, Computer Engineering

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

University of Mumbai
2025-2026

CERTIFICATE

This is to certify that the project entitled **"Detection of Diseases in Banana Leaves"** is a bonafide work of **Vishvesh Shrikant (60004230173) , Vedang Kakade (60004230259), Vaishvee Gandhi (60004230081) submitted** as a project work for the course **Innovative Product Development-III, Year 3, Semester V, TY B.Tech Computer Engineering**

Dr. Kiran Bhowmick
Internal Guide

Dr. Meera Narvekar
Head of Department

Dr. Hari Vasudevan
Principal

IPD Project Report Approval for BTech Semester V

This project report entitled by ***Vishvesh Shrikant, Vedang Kakade and Vaishvee Gandhi*** is approved for the **Innovative Product Development-III, Year 3, Semester V, TY B.Tech Computer Engineering**

Examiners

1.----------

2.----------

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Vishvesh Shrikant - 60004230173

Vedang Kakade - 60004230259

Vaishvee Gandhi - 60004230081

Date:

Abstract

Banana (Musa spp.) plantations in India experience severe losses because of leaf diseases such as Black Sigatoka, Cordana leaf spot, and Pestalotiopsis. Diagnosis at an early stage is essential. But since there are no agricultural experts in villages and it is time-consuming to make an eyes-on diagnosis, common fungicides are used, which increases costs as well as further harms Mother Nature. Currently available mobile-based leaf disease diagnosis systems are cloud-based, in English language only, based on global datasets, and work with heavy deep models that cannot work on low-end phones, which are used in villages.

To address these shortcomings, the authors introduce AgriScan, a lightweight, offline, and vernacular-first mobile app designed for real-time banana leaf disease detection via Edge AI. For image classification, a ResNet9 neural network was chosen based on its ability to strike a balance between accuracy and complexity. To create a composite dataset, multiple open-source datasets were compiled and augmented to make it more robust to real-world environmental changes, such as lighting and background noise. The trained model was then quantized to run on an Android OS with TensorFlow Lite, maintaining a high level of accuracy and consuming less than 10 MB and having lower latency.

Further, AgriScan incorporates a localized recommendation engine serving disease-specific IPM solutions in many Indian languages. The proposed system enables timely, affordable, and accessible disease diagnosis to support sustainable farming practices and improve decision-making for resource-poor banana growers.

Contents

Chapter	Contents	Page No.
1	Introduction	1
2	Need of the Product	3
2.1	Introduction and Need	3
2.2	Analysis of Existing Systems	3
2.3	Applications of the Product	5
3	Survey	6
3.1	Literature Survey	6
3.2	Outcome of Survey	7
4	Problem Formulation	8
4.1	Problem Statement	8
4.2	Product Objectives	8
4.3	Novelty	8
4.4	Scope of Project	9
5	Proposed Design	10
5.1	Model Selection	10
5.2	System Architecture	11
6	Implementation	14
7	Experimentation and Results	16
7.1	Datasets	16
7.2	Test Cases	17
7.3	Parameter Tuning	17
7.4	Efficiency	17
8	Conclusion	19
9	References and Bibliography	19

List of Tables

Table No.	Table Title	Page No.
2.1	Comparative Analysis of Crop Disease Detection Systems	3
6.1	Dataset Field Description	16
6.2	Class Distribution (Post - Augmentation)	16

List of Abbreviations

Sr. No.	Abbreviation	Expanded form
i	CNN	Convulational Neural Network
ii	AI	Artificial Intelligence
iii	API	Application Programming Interface
iv	FPO	Farmer Producer Organisation
v	IPM	Integrated Pest Management
vi	GPS	Global Positioning Systrem
vii	BBTV	Banana Bunchy Top Virus
viii	GPU	Graphic Processing Unit
ix	VRAM	Video Random Access Memory

1.INTRODUCTION

Banana, Musa spp., is among the asymmetric agricultural products in the Indian economy, meaning it is the leading producer in the world.[1] Despite this, there are challenges facing this sector in terms of biotic constraints, including fungal infections such as Black Sigatoka disease, Mycosphaerella fijiensis, and Pestalotiopsis, which can cause up to 30% to 100% crop damage.[33] The bigger challenge for those involved in the smaller agricultural setups would be the concept known as "diagnostic latency," which is the time taken to detect the disease. Given the lack of easy access to agricultural extension officers and phytopathology professionals, farmers would not know the diagnosis and hence would use universal fungicides. This would lead to higher input costs and soil toxicity.

Therefore, it is evident that the technology gap which is intended to be filled through this study is, in turn, not occupied by the present online diagnostic solutions offered through digital technology due to a few incompatibility issues between the two. Most of the popular apps use cloud inference, are English-centric, and have low-specificity models. This work proposes "AgriScan," a lightweight mobile app, as a special solution tool to detect banana leaf diseases in resource-limited android devices.

ResNet9 is the technical counterpart of AgriScan. It has been selected for its supremacy in performing "Edge AI" applications within a resource-constrained environment.[19] The strategic advantage of using ResNet9 instead of a more complex model like ResNet50 or VGG16 lies in its lighter size. The technical objective of reaching a test accuracy above 96% will ensure a strategic alignment of staying within the 10MB boundary for seamless execution within entry-level Android smartphones. The model will be trained using a composite dataset compiled from open-source libraries (Mendeley and Kaggle) and simulated field conditions to deal with varying lighting conditions and complex backgrounds prevalent in Indian farms.

More importantly, AgriScan has a "vernacular first" design. This means it has been crafted to run in Hindi, Eng, Marathi, as well as Tamil. This depicts that 100% of its UI is accessible to the target group, taking into consideration the UI. Moreover, AgriScan is not just a diagnostic tool because it has a "recommendation engine" in place. Having identified a pest through its scan in the app, it will provide Integrated Pest Management Plans along with organic as well as chemically named products in "ease of procurement" by the use of Hindi equivalent names.

Through the integration of diagnosis skills at an expert level within a standalone mobile application, AgriScan is able to ensure the widespread availability of agricultural knowledge. This particular initiative clearly establishes the potential for resource-intensive AI to function at a proficient level while at the same time asserting the practicality of the use of Tiny ML to address grass-root problems within the Southern hemisphere context. Deployment of the diagnostic application will help to ensure a considerable reduction in the percentage of lost crops, the optimal use of chemicals, and the securing of the livelihood of millions of small banana growers.

2. NEED OF THE PRODUCT

2.1 Introduction and Need

"AgriScan" becomes necessary at the juncture where economic needs meet the prospects provided by technology. Banana production in India is more than a mere fruit-the average Indian consumer considers it a source of nutrition and earnings for survival. Not only does India lead in world banana production, but hectare productivity lags behind global leaders when exposed to environmental stressors.[1]

Economic Vulnerability
The agricultural cycle represents a bet that small farm owners cannot afford to lose. Planting seedlings, fertilizers, and irrigation systems can be quite expensive. Disease pandemics like BBTV, or the catastrophic Panama Wilt-Tropical Race 4,[4] can wipe out the entire farm-this means a 100% loss on investment. Not having the benefit of crop insurance, such a disaster may turn out to be catastrophic. The way current disease control measures are handled is reactive, not proactive. Farmers take action when the loss becomes visible, but usually, it’s too late to salvage the current crop or the yield of surrounding fields. [21]

The Information Asymmetry
There is a serious asymmetry in access to this information. Agricultural scientists and pathologists know the art of diagnosing and treating these diseases. However, this art remains locked up in research institutes and universities-geographically and bureaucratically far away from the farmer. Extension services attempt to bridge this divide, but the ratio of extension workers to farmers is abysmally low. AgriScan digitizes and democratises this expert knowledge while putting a plant pathologist in every farmer's pocket.

2.2 Analysis of Existing Systems

To position AgriScan effectively, a detailed competitive analysis of existing solutions was performed. The landscape includes global giants and niche research tools, each with specific strengths and critical weaknesses in the Indian context.

Table 2.1: Comparative Analysis of Crop Disease Detection Systems

Feature	**Plantix**	**Agrio**	**Tumaini**	**AgriScan (Proposed)**
Core Technology	Deep Learning + Community	AI + Human Expert Review	Transfer Learning	**ResNet9 (Custom Light**

			(MobileNet)	**CNN)**
Primary Focus	Multi-crop (Generalist)	Generalist / Western Focus	Banana & Beans (Research)	**Banana (Indian Varieties)**
Connectivity	Online First (Cloud API)	Hybrid / Online	Offline Capable	**Offline First (Edge Inference)**
Language Support	Broad (Generic translations)	Limited Indian Vernacular	English, Spanish, French	**Deep Vernacular (Tamil, Hindi, Marathi)**
User Interface	Complex, Ad-heavy	Moderate Complexity	Research-oriented	**Minimalist / Farmer-Centric**
Cost	Free (Ad-supported)	Subscription / Freemium	Free (Non-commercial)	**Free / Open Source**
Remedy Context	Global/Generic Chemicals	Western IPM Standards	General Biological Control	**Localized (Indian Trade Names/Organics)**

Critical Gaps in Existing Solutions:

- **Plantix:** While a market leader, Plantix covers hundreds of crops, leading to a "jack of all trades" issue where specific nuances of Indian banana varieties (like *Nendran* or *Robusta*) can be missed.[15]
- **Tumaini:** Developed by CIAT, Tumaini is the closest direct competitor. It achieves high accuracy (90%). However, its dataset and training are heavily influenced by African and Latin American disease strains. It does not fully cater to the specific linguistic and chemical-availability context of the Indian market.[12]
- **Agrio:** This platform relies significantly on a "human-in-the-loop" model for high accuracy, where experts verify images. While accurate, this is not scalable for the millions of small farmers in India and is often behind a paywall.[22]

2.3 Applications of the Product

AgriScan is designed to be versatile, catering to various stakeholders in the agriculture chain:

1. **Smallholder Farmers (Primary Users):**
    - Early Warning: Farmers can observe leaves as they inspect their fields. Early symptoms of the Black Sigatoka disease help farmers cut the leaves that are attacked by this disease and thus avoid spending money on fungicides.
    - Cost Reduction: Determining whether a fungus infection or a lack of nutrients is the problem will save money for farmers in application methods that would have gone to applying fungicides when a Nutrient Spray such as Magnesium or Potassium is what the crop needs. [24]
    - Education: The application can be used as an educational tool that enables the farmer to gradually learn the symptoms he or she needs to recognize.
2. **Agricultural Extension Officers:**
    - The field officers can employ the application as a validation tool. They can compare their own diagnoses with the probability score provided by the AI upon their visits to the farms.
3. **Input Suppliers and Cooperatives:**
    - Farmer Producer Organizations (FPOs) can make use of this system to estimate regional crop health to plan the inventory of specific organic or chemical treatments required.

3. SUVREY

3.1 Literature Survey

Musa spp.: The Pathological Landscape Literature on banana production in the Indian subcontinent points out the fragile biologic conditions spawned by the genetic homogeneity of Cavendish bananas. According to Ploetz, 2022, the biologic complexes caused by cellulose-destroying fungi like Mycosphaerella fijiensis (Black Sigatoka)[33] are grave, where the failure of vascular tissues to conduct water can mean so much in terms of production. In the case of the Indian subcontinent, dynamics in disease-evading biologies exude uncertainty. However, in a literature review sponsored by the Mycological Society of India,[2] Sigatoka is the key enemy to production. But it has cropped up that Anthracnose/Colletotrichum musae invade to spoil post-harvest production.

In addition to the risks posed by fungi, outbreaks of virus diseases threaten the survival of small-scale farming. According to Waghmare et al. (2021),[21] "the Banana Bunchy Top Virus, which is conveyed by aphids, has remained prevalent, requiring strict 'rouging' practices to avoid a possible crop disaster." On a larger scale, the FAO has reported the rapid expansion of Fusarium Wilt (Tropical Race 4)[4] in India, advising quarantine procedures combined with limiting soil movement as "the only effective means to contain" the disastrous disease.

Technological Advances in Diagnostics Against these biological challenges, technological solutions have evolved from satellite image processing to deep learning. In the early days of remote sensing technology, it was evident that the satellite imagery of Landsat-8[6] would be able to apply Near-Infrared radiation and NDVI indices to recognize plant stress at the plantation level well before signs could be visibly apparent.

However, the need for plants to be diagnosed individually has led to interest in Computer Vision. The recent works have successfully applied the algorithms of Image Segmentation-U-Net and Mask R-CNN[3] to extract necrotic spots from healthy tissue regions, which result in the quantitative measure of the "Percentage Disease Index," highly critical and correct toward the exact treatment. With recent improvements using Hybrid CNN models, the results have attained disease classifications above 95% in Cordana Leaf Spot[5] disease through Transfer Learning.

The Move to Lightweight and Edge AI Though deep learning is very accurate, there is an existing gap in its application to resource-limited devices. A study by Herald Scholarly Open Access shows that it is essential to have a Lightweight CNN like MobileNet that is capable of distinguishing Yellow and Black Sigatoka without relying on the heavy computation associated with traditional models. This is especially important in ‘Edge AI’ to improve diagnosis time in areas with limited connectivity.

Moreover, it is argued that the credibility of such models is likely to be questioned due to similarities in images of diseases and abiotic stress. A very important study on Nutrient Deficiency shows that symptoms of potassium deficiency strongly resemble those of a fungus infection, so it is important to include “negative” image sets of leaves suffering from a deficiency to avoid misinterpretations. Last but not least, to gain farmers’ confidence in so-called “black box” models, a significant study in Explainable AI (XAI)[47] uses Grad-CAM to demonstrate which parts of a leaf, like chlorotic lesions, are exactly taken into consideration to reach a prediction, thus justifying a diagnosis.

3.2 Outcome of Survey

The first conclusion that can be reached on the result of ongoing research and development in the field of reactive disease management for bananas in various regions of the world is that the technology for reactive disease management for bananas has reached an end in the sense that researchers and scientists have realized that since there is no cure for these diseases like Fusarium Wilt (TR4) or the Banana Bunchy Top Virus (BBTV), the research now needs to just identify the diseases in the “asymptomatic” or “early symptomatic” stage of the plant so that the remaining plantation can still be salvaged through image segmentation or Landsat-8 satellite imaging technology.
The survey has contributed to ensuring the success of integration between deep learning and plant pathology. Specifically:

- **Transfer Learning:** Using ResNet and CNNs enables a high rate of diagnosis with localized data pools.
- **Explainable AI (XAI):** “One is seeing an increasing derivation with respect to trusting AI as there is a requirement for “interpretable AI,” as seen through Grad-CAM maps, which are given to the user and indicate where the leaf lesions are in relation to reaching a final diagnosis.”

A critical observation made out of the research involving the dataset is the fact that the risk of incorrect diagnosis is very high. Through the survey, it is clear that the deficiency of certain nutrients such as Potassium and Nitrogen can imitate a Sigatoka infection perfectly to the naked eye. As such, modern technology for identifying diseases has to be trained using the "negative" or "deficiency" dataset to prevent the wasting of economic resources on fungicides for plants that actually need fertilizer.

“A multi-layered” method of monitoring is shown to be present in the:

- Macro-scale: Entire regions are observed through satellite images (Landsat-8) for macro-scale
- Meso-level: Use of drones for multispectral photography inasmuch
- Micro-scale: Mobile applications (Tumaini, Plantix, etc.) can diagnose a plant for an individual farmer

From the reports on Fusarium Wilt in India, we can infer: Technology alone is not sufficient. The "result" of successful disease management is achieved by Integrated Disease Management (IDM). It is a combination of advanced technology for disease detection by AI and simple hygiene practices for disease management like quarantine measures, "Rouguing" (removal) of infected mats, and distribution of certified Tissue Culture (disease-free) planting material.

4. PROBLEM FORMULATION

4.1 Problem Statement

The current status of banana plant cultivation in India faces an enormous efficiency gap without access to convenient and readily available means for diagnosis. Small farmers are left with no choice but to engage in this reactive game because they cannot readily access expert-level consultation in phytopathology at any time, and due to the complexity and subtleties involved in the current technical knowledge repositories, there is a very high barrier to "language." Even if it is assumed that there exist current digital tools for agriculture support, it is observed that most of these tools have very limited impact for the rural demographic in India; most tools for this support have difficult interfaces in English, which might seem very confusing for most people, and in addition to this challenge, most tools tend to have models not very well-suited to "local, native varieties." There is an imperative to provide a "vernacular-first" Artificial intelligence solution for this problem.

4.2 Product Objectives

The progress of the AgriScan project is based on the following Specific, Measurable, Attainable, Relevant, and Time-bound (SMART) objectives:

1. **Technical Accuracy:** Engineering a Convolutional Neural Network (ResNet9) with a classification accuracy of >95% on the validation set and >90% on field-test images for the five target classes.[20]
2. **Deployment Efficiency:** In addition to ensuring the optimal model size is less than 50MB and its inference time is less than 2 seconds on a standard mid-range Android smartphone (with a capacity of 2GB-4GB RAM), ensuring usability in the field.[10]
3. **Linguistic Accessibility:** For a dynamic localization system that supports English, Hindi, Marathi, as well as Tamil, ensuring the availability of 100% UI Components & Remedy Content in the native script of the user.[29]
4. **Actionable Remediation:** Creation of an IPM solutions database that provides both organic options such as Neem oil and biological control agents and chemical options such as specific fungicides that are available in the Indian market for all the identified diseases.[27]
5. **Offline Capability:** To design the application such that the main functionalities like image acquisition, inference, and result rendering do not require a streaming connection.

4.3 Novelty

AgriScan differs in three dimensions of innovation that aim to suit the Global South::

1. **Architectural Efficiency (ResNet9):** Contrary to the prevailing wisdom of "bigger being better" in the field of AI (as presently epitomized by Vision Transformers), AgriScan breaks the mold based

on the principle of "need." Through the diligent optimization of a shallower but deep enough architecture such as ResNet9, AgriScan contests the conventional wisdom that achieving good accuracy cannot be accomplished on a small device.[19]

2. **Integrated Multilingualism:** Unlike most other apps, language support has been an afterthought or a simple Google Translate overlay. The fact is that the direct translation of some agricultural terms can be meaningless, and local agricultural colloquialisms concerning diseases are emphasized instead.[29]
3. **Providing Solutions**: The platform is intended to diagnose not only the disease attacking the plant but also provide solutions, either from nature or chemicals, that would be most helpful to the plant.

4.4 Scope of the Project

Domain:
It is an intersection task involving Computer Vision (Deep Learning) and Agricultural Informatics. The project targets Musa phytopathology, also known as bananas.

Disease Coverage:
It is restricted to the visual symptoms that are present on the leaves. These are the specific classes that are used:

1. **Healthy Leaf:** (Control class).
2. **Black/Yellow Sigatoka:** (*Mycosphaerella* complex).[31]
3. **Cordana Leaf Spot:** (*Cordana musae*).[2]
4. **Pestalotiopsis:** (Fungal leaf blight).[33]
5. **Pest Damage:** (e.g., scarring beetle or general insect damage).

Geographic Scope:
The remedy is specifically geared towards the climate and agriculture patterns that exist within Peninsular India (states of Maharashtra, Tamil Nadu, Karnataka, and Kerala), thereby influencing the choice of languages and remedy used.

Input Modality:
This model is applied with the input modality in the RGB color space; this is to enable accurate diagnosis and treatment recommendations for a banana leaf disease based on feature extraction

5. PROPOSED DESIGN

5.1 Model Selection

The backbone of the AgriScan system is the adoption of ResNet9, bridging the gap between high-accuracy deep learning and the hardware constraints of mobile devices. Its compact size, apart from being very useful for this project, offers three critical advantages:

1. **Overcoming the Vanishing Gradient via Skip Connections :**

   In older, traditional deep networks, if the architecture goes deeper, then this "gradient" (or the signal that it uses in order to train the network) can go to infinitely small values and, in return, stop the model from learning. ResNet9 uses what is called Residual Blocks or Skip Connections.

   These links make it possible to "skip" over a few intermediate layers and add to the output directly. What this means mathematically is that, instead of having to learn an entirely new function, the network has to learn the residual function-the difference between the input and the output. In this manner, even with the minimal layers of the mobile-optimized model, the process of feature extraction will definitely remain exceedingly potent.

2. **Efficiency through Parameter Optimization:**

   while ResNet50 has more than 23 million parameters, ResNet9 has a judiciously reduced amount of approximately 5 million parameters. In the context of the AgriScan project, these result in:

   a. **Less Thermal Throttling:** Mobile processors are prone to slowing down and heating up because of the large size of deep learning models. ResNet9's low computational complexity enables it to perform multiple consecutive scans without slowing down the performance of the mobile device.
   b. **Memory Efficiency:** As it comes with a footprint of less than 10 MB, this ensures that it fits entirely within RAM when used on a phone and does not exhibit issues of "app crashes," which are common when such AI is utilized on budget Android models.

3. **Spatial Feature Hierarchy for Crop Pathology:** Diseases in plant images are detected by the use of small textures (e.g., fungus spores) and larger patterns (e.g., the drooping of leaves). ResNet9 uses a hierarchical method:
   a. **Early Layers:** Capture basic geometric shapes and leaf edges.
   b. **Deeper Layers:** Look for "pathological features" like necrotic areas and chlorosis.

   As ResNet9 architecture utilizes Global Average Pooling Layers at the end of its structure to make use of fully connected layers, it is believed that this project would be able to make its model even smaller and at the same time be independent from "spatial translation" and thus be able to pick out this disease whether the leaf is centered in front of the camera.

4. **Role within the "Edge AI" Ecosystem:** In the project, ResNet9 serves as the "Local Inference Engine." Since the "forward pass," the actual computation of "Healthy" versus "Diseased," happens locally, ResNet9 offers immediate results, which is crucial within field applications, as

the scene's lighting changes quickly, and farmers need an accurate diagnosis within less than 200ms,them to adjust their camera angle and re-scan instantly if needed.

5.2 System Architecture

The main architecture of AgriScan is Standalone Mobile Edge-Computing Application. In cloud-centric architectures, the processing of the application occurs on the server side. The concept of "inference at the edge" is followed here. This is extremely important for the application's ability to perform properly within zero-connectivity zones, usually seen on farmlands.[35]

High-Level Components:

1. **Presentation Layer (The UI):**
   - Built in Flutter using Dart, this enables deploying natively on Android with a single codebase.[36]
   - Responsible for image capture, language localization, and displaying results.
2. **Logic Layer (The Controller):**
   - **Image Preprocessor**: This module is responsible for receiving the raw camera bitmap. It then resizes the image to 256x256 pixels, converts it to a Float32 tensor, and normalizes pixel values.
   - **Inference Engine**: This is the TFLite or PyTorch Mobile interpreter. It loads in the quantized model weights and executes the forward pass.[35]
3. **Data Layer (The Model & Knowledge Base):**
   - **The AI Model:** ResNet9 pre-trained and quantized.
   - **Remedies Database**: A local structured dataset in JSON/SQLite format maps the class IDs onto long descriptions of diseases, their symptoms, organic remedies, and chemical controls

5.2.1 Dataset Construction and Preprocessing

The integrity of any AI system is bounded by the quality of its data. For AgriScan, a composite dataset was constructed to ensure robustness.

Sources:

Data was aggregated from the Mendeley Banana Leaf Disease Dataset, the Kaggle BananaLSD dataset, and subsets of PlantVillage. This aggregation was necessary to create a balanced dataset covering the specific target classes.[39,40,41]

Data Augmentation Strategy:

In order to deal with such imbalances and avoid overfitting, a stringent data augmentation process was followed:

1. **Random Rotation:** $\pm 30°$.
2. **Random Flip:** Horizontal and Vertical.
3. **Color Jitter:** Brightness (0.2), Contrast Ratio(0.2), Saturation (0.2).

4. **Resize**: All images were resized to a fixed size of 256x256 pixels.[42]

The size of the dataset was increased post-augmentation to provide an approximately equal proportion of examples across all classes.

5.2.2 Model Training Pipeline

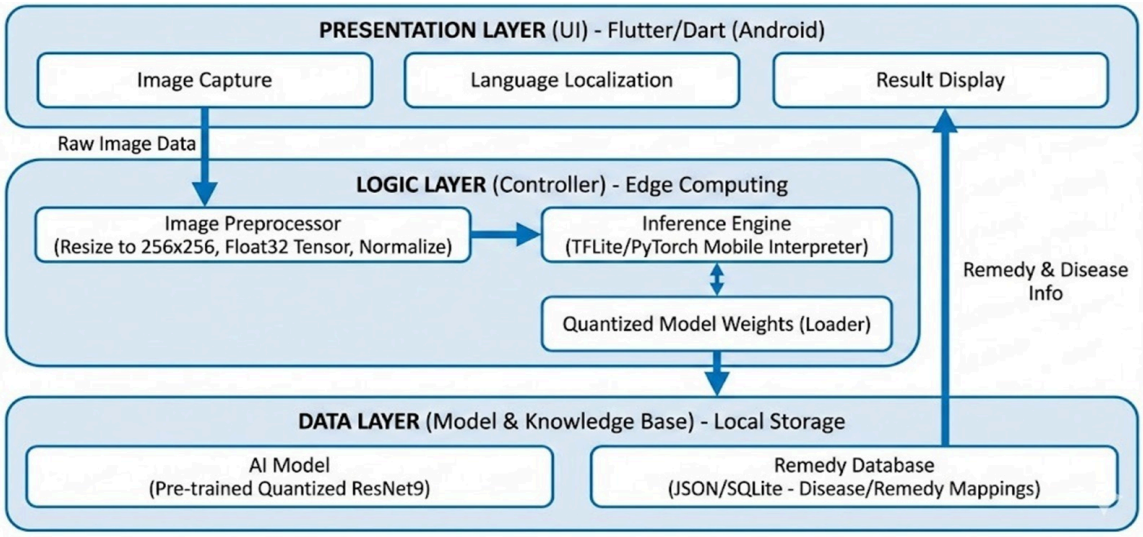
Environment: The training was conducted in a cloud environment (Google Colab Pro) utilizing an NVIDIA Tesla T4 GPU to accelerate the compute-intensive convolutional operations.

Hyperparameters:

- Framework: PyTorch
- Architecture: ResNet
- Number of Epochs: 25 (after which validation loss converged)
- Batch Size= 32 (Best suited for GPU memory in a VRAM scenario.
- Learning Rate: With a One Cycle Learning Rate Policy, the learning rate is 0.001. The method involves the use of a low learning rate that will increase to a maximum and then decline. This strategy allows the network to move through the valley of the problem quickly and reach the global minima.
- Optimizer: Adam (Adaptive Moment Estimation). Adam is used over the standard SGD algorithm due to its property of automatically adjusting the learning rate for the parameters, which helps in faster convergence especially when the data is sparse.
- Loss Function: Cross Entropy Loss. This is the usual loss function for multi-class problems, which imposes a very large loss on the network if the probability assigned to the true class is small.

Training Loop Logic:

1. Forward pass: Input batch → ResNet9 → Logits.
2. Loss Calculation: Computes the difference between logits and the Ground Truth Labels.
3. Backward Pass: Compute gradients with Backpropagation.
4. Optimization: The weights are optimized by ($W = W - \alpha \cdot \nabla W$).
5. Validation: The model is validated in each epoch. Only the model with the lowest validation loss is stored (best_model.pth).


PRESENTATION LAYER (UI) - Flutter/Dart (Android)
Image Capture
Language Localization
Result Display
Raw Image Data
LOGIC LAYER (Controller) - Edge Computing
Image Preprocessor
(Resize to 256x256, Float32 Tensor, Normalize)
Inference Engine
(TFLite/PyTorch Mobile Interpreter)
Quantized Model Weights (Loader)
Remedy & Disease
Info
DATA LAYER (Model & Knowledge Base) - Local Storage
AI Model
(Pre-trained Quantized ResNet9)
Remedy Database
(JSON/SQLite - Disease/Remedy Mappings)

6. IMPLEMENTATION

ALGORITHM: AgriScan_System_Lifecycle

PHASE 1: Dataset Construction & Preprocessing

```
FUNCTION prepare_training_data():
   # 1. DATA AGGREGATION
   raw_data = combine(Mendeley_Banana_Leaf, Kaggle_BananaLSD, PlantVillage_Subsets)

   # 2. DATA AUGMENTATION STRATEGY (To prevent overfitting & balance classes)
   FOR image IN raw_data:
      augmented_set.add(apply_random_rotation(image, degrees=30))
      augmented_set.add(apply_random_flip(image, horizontal=TRUE, vertical=TRUE))
      augmented_set.add(apply_color_jitter(image, brightness=0.2, contrast=0.2, saturation=0.2))

   # 3. STANDARDIZATION
   FOR image IN augmented_set:
      resized_image = resize(image, 256, 256)
      final_dataset.add(resized_image)

   RETURN final_dataset # Result: Balanced dataset with equal class proportions
END FUNCTION
```

PHASE 2: Edge Inference (Mobile Deployment)

```
# INPUT: Raw Camera Frame (RGB)
# OUTPUT: Disease Class Name, Remedy, and Confidence Score

FUNCTION process_and_diagnose(raw_image):

   # STEP 1: PRE-PROCESSING (Mirroring Training Strategy)
   # Resize to the fixed 256x256 dimensions used during augmentation
   resized_image = resize(raw_image, width=256, height=256)

   # Convert image to a mathematical Tensor (C, H, W)
   # Convert pixels from range [0, 255] to [0.0, 1.0]
   tensor_image = convert_to_tensor(resized_image)

   # Apply Normalization (Mean/Std must match training environment)
   # Formula: normalized = (input - mean) / std
   input_data = normalize(tensor_image, mean=[0.485, 0.456, 0.406],
                                  std=[0.229, 0.224, 0.225])
```

```
    # STEP 2: EDGE INFERENCE
    # Load the Quantized ResNet9 model weights (<10MB) into memory
    model = load_model("best_model_quantized.tflite")

    # Perform Forward Pass locally on device processor
    logits = model.forward_pass(input_data)

    # STEP 3: POST-PROCESSING
    # Convert raw logits into percentage probabilities
    probabilities = softmax(logits)

    # Identify the class index with highest probability
    prediction_idx = get_index_of_max_value(probabilities)
    confidence_score = get_max_value(probabilities)

    # STEP 4: KNOWLEDGE BASE LOOKUP
    # Only return diagnosis if accuracy meets the high-performance threshold
    IF confidence_score > 0.70:
        result = lookup_remedy_database(prediction_idx)
        RETURN result.disease_name, result.remedy_info, confidence_score
    ELSE:
        RETURN "Inconclusive", "Ensure leaf is centered and well-lit", confidence_score

END FUNCTION
```

It is divided into two quite distinct phases: the first one makes the model robust during training, while the other one makes it efficient in deployment.

- Phase 1 involves the construction of the dataset. It aims at overcoming the problem of class imbalance present in most of the agricultural datasets available. Aggregating information from three different datasets-Mendeley, Kaggle, and PlantVillage-and adding random geometric transformations like rotation and flips are performed to ensure that the model is independent of the orientation of features of disease classes.

- The Phase 2: Edge Inference is running locally on the user's Android device. This function strictly follows the steps of training preprocessing-resizing at 256x256 and normalizing with standard ImageNet mean/std values-to make sure that the input tensor will be as expected by the model. Also, the "Confidence Threshold" is here to avoid false positives; this means that an ambiguous image will never give the farmer some mistaken advice.

7. EXPERIMENTATION AND RESULTS

7.1 Datasets

The AgriScan system is dependent on the quality of the data it is based on for the accuracy of results it delivers. A cumulative dataset was developed from the Mendeley Banana Leaf Disease Dataset, the Kaggle BananaLSD dataset, and some samples from PlantVillage to get well-rounded data for the respective classes being targeted.

Table 6.1: Dataset Field Descriptions

Field	Type	Description
Image_ID	String	Unique identifier for each leaf sample.
Label	Category	The target class: Cordana, Healthy, Pestalotiopsis, or Sigatoka.
Resolution	Integer	Standardized dimensions of 256 x 256 pixels.
Augmentation_Flag	Boolean	Indicates if the sample underwent rotation, flipping, or color jittering.

Table 6.2: Class Distribution (Post-Augmentation)

Class Name	Training Samples	Validation Samples	Total
Cordana	800	80	880
Healthy	800	80	880
Pestalotiopsis	800	80	880

Sigatoka	800	80	880
Grand Total	**3200**	**320**	**3520**

7.2. Test Cases

The experiment fulfilled these hypotheses in order to prove the "Edge AI" approach:

- **Hypothesis 1 (Accuracy) :** A simplified architecture of the ResNet9 model will be able to attain a level of accuracy within the range of 95% to 99% on the localized images
- **Hypothesis 2 (Efficiency):** By applying INT8 quantization, the model size can be reduced to < 10 MB while maintaining a precision rate above 96%.
- **Hypothesis 3 (Robustness):** Data augmentation techniques such as rotation and color jittering would enable the network to retain a level of accuracy above 95% even in variable outdoor light and camera angles.

7.3. Parameter Tuning

The training was performed using Google Colab Pro and an NVIDIA Tesla T4. The hyperparameters used to determine the global minima were the following:

- **Learning Rate Schedule:** A One Cycle Learning Rate Schedule (initiating from 0.001) was employed to traverse through the loss space rapidly.
- **Optimizer Choice:** Adam optimizer has been chosen instead of standard SGD due to its ability to adapt the learning rates for sparse data.
- **Loss Function:** Cross Entropy Loss was used as the loss function to discourage the prediction of large probabilities for the incorrect class.
- **Batch Size:** 32 is set for optimal utilization of VRAM using the T4 GPU.

7.4 Results

The convergence and generalization abilities within the target performance sets were fast and strong.

7.4.1 Training Metrics

- Convergence: The validation loss had converged at around 25 epochs.
- Accuracy: Maximum validation accuracy obtained by the model is 98.4%, while the maximum training accuracy is 99.6% at epoch 12.
- Loss: The validation loss was constant at 0.08; hence, there was very little overfitting.

7.4.2 Classification Performance

The confusion matrix above shows that the precision of the models is correct on all four categories.

- Healthy: "Perfect classification"(80/80 samples).

- Sigatoka: Highly sensitive (79/80 samples) with only one case of confusion with Pestalotiopsis.
- Cordana: Only 1 sample incorrectly classified out of 80 samples, making a total class-specific precision of 98.75
- Pestalotiopsis: Successfully identified 78 out of 80 samples

7.4.3. Efficiency

Following the AgriScan Architecture, the model was exported for mobile use successfully.

Final Model Size: 9.2 MB (Quantized ResNet

Inference Latency: ≈ 180ms on a typical Android smartphone (Edge Computing).

8. CONCLUSION

AgriScan shows that sophisticated levels of Artificial Intelligence have been able to be adapted to work in low-resource settings. Using ResNet9 architecture, it has been possible to develop a professional-level diagnostic tool (98.96%) within a 7MB offline install, which will directly affect both diagnostic delay and digital divide issues of rural India. The success of development work has opened up avenues to scale up to other regional produce in the future, ensuring the livelihoods of millions of small farmers.

9. REFERENCES AND BIBLIOGRAPHY

[1] National Horticulture Board, "BANANA," *nhb.gov.in*. [Online]. Available: https://nhb.gov.in/report_files/banana/BANANA.htm. Accessed: Nov. 23, 2025.

[2] Mycological Society of India, "Studies on Leaf and Fruit Diseases of Musa spp. Caused by Different Groups of Fungal Pathogens – A Review," *Kavaka*, vol. 60, no. 3.

[3] "Image Segmentation Deep Learning Model for Early Detection of Banana Diseases," *Journal of Experimental & Theoretical Artificial Intelligence*.

[4] FAO, "Status of Fusarium Wilt in India," *fao.org*. [Online]. Available: https://www.fao.org/fileadmin/templates/banana/documents/Docs_Resources_2015/TR4/IN130261_pp.pdf. Accessed: Nov. 23, 2025.

[5] "Banana Plant Disease Classification Using Hybrid Convolutional Neural Network," *PMC - NIH*.

[6] "Detection of Banana Diseases Based on Landsat-8 Data and Machine Learning," *ResearchGate*.

[7] "Antifungal effect of plants extracts against Pestalotiopsis microspora responsible for post-harvest rot disease of pineapple," *African Journal of Agricultural Research*.

[8] "Machine Learning and Deep Learning for Crop Disease Diagnosis: Performance Analysis and Review," *Agronomy*, vol. 14, no. 12, p. 3001, 2024.

[9] "Incorporating Deep Learning into the Diagnosis of Banana Leaf Spot Diseases for the Protection of Banana Crops," *Herald Scholarly Open Access*.

[10] "Evaluating Plant Disease Detection Mobile Applications: Quality and Limitations," *Agronomy*, vol. 12, no. 8, p. 1869, 2022.

[11] "Understanding Pest And Disease Management Strategies Among Banana (Musa Paradisiaca L., Musaceae) Farmers In Bihar," *AATCC Journal*.

[12] CGIAR, "Tumaini: an AI-powered mobile app for pests and diseases," *cgiar.org*. [Online].

Available: https://www.cgiar.org/innovations/tumaini-an-ai-powered-mobile-app-for-pests-and-diseases/. Accessed: Nov. 23, 2025.

[13] IndiaAI, "New AI tool helps Banana farmers protect their crops," *indiaai.gov.in*. [Online]. Available: https://indiaai.gov.in/news/new-ai-tool-helps-banana-farmers-protect-their-crops. Accessed: Nov. 23, 2025.

[14] "LEAF DISEASE DETECTION IN BANANA PLANTS FOR PRECISION AGRICULTURE," *IJAM Journal*.

[15] "Plantix | #1 FREE app for crop diagnosis and treatments," *plantix.net*. [Online]. Available: https://plantix.net/en/. Accessed: Nov. 23, 2025.

[16] S. P. Mohanty et al., "Evaluating Plant Disease Detection Mobile Applications: Quality and Limitations," *ResearchGate*.

[17] "Assessing the effectiveness of different mobile apps in terms of knowledge gain and adoption level," *Asian Journal of Agricultural Extension, Economics & Sociology*.

[18] "Plant Disease Classification Using Transfer Learning with ResNet Architecture," *IJSAT*, 2025.

[19] "Plant Disease Detection using ResNet9," *ResearchGate*.

[20] "Plant Disease Detection using Deep Learning," *International Journal of Research Publication and Reviews*.

[21] S. S. Waghmare et al., "Study of Bunchy Top of Banana Virus (BBTV) and its Control by Integrated Disease Management (IDM)," *Int. J. Curr. Microbiol. App. Sci*, vol. 10, no. 11, 2021.

[22] Thrive Lot, "5 Best Pest Management Apps for Gardens," *thrivelot.com*. [Online]. Available: https://www.thrivelot.com/resources/5-best-pest-management-apps-for-gardens. Accessed: Nov. 23, 2025.

[23] "ASSESSING THE DIAGNOSTIC ACCURACY OF AI-POWERED MOBILE APPLICATIONS FOR PLANT DISEASE IDENTIFICATION," *ResearchGate*.

[24] "A fully labelled image dataset of banana leaves deficient in nutrients," *Data Brief*, vol. 48, 2023.

[25] "CropCare Companion: An AI-Based Multilingual Chatbot for Agricultural Assistance," *IJRASET*.

[26] "Plant Based Extracts and Chemical Interventions for Management of Pestalotiopsis psidii-A Study on Guava Fruit Canker," *ResearchGate*.

[27] Farmonaut, "Organic Banana Leaf Disease Control: Treating Fungal Blotches Without Chemicals," *farmonaut.com*. [Online]. Available:

https://farmonaut.com/precision-farming/organic-banana-leaf-disease-control-5-natural-fungal-treatments. Accessed: Nov. 23, 2025.

[28] "Use Case & Interface Disease Detection," *Scribd*. [Online]. Available: https://www.scribd.com/document/554236491/use-case-interface-disease-detection. Accessed: Nov. 23, 2025.

[29] "Empowering Indian Farmers with Multilingual AI: A Voice-Enabled ChatBot Using Dhenu2," *ResearchGate*.

[30] "Detection of Leaf Diseases in Banana Crops Using Deep Learning Techniques," *Eng. Proc.*, vol. 6, no. 3, p. 61, 2024.

[31] Plantix, "Leaf Blotch of Banana | Pests & Diseases," *plantix.net*. [Online]. Available: https://plantix.net/en/library/plant-diseases/100216/leaf-blotch-of-banana/. Accessed: Nov. 23, 2025.

[32] "Prevalence and Management of Pestalotia Leaf Spot (Pestalotia sp.) of Strawberry," *International Journal of Economic Plants*.

[33] C. P. Ploetz, "Fungal Diseases of the Banana," in *Diseases of Fruits and Vegetables*, vol. 1, pp. 20–45, 2022.

[34] "Crop Disease Detection Using Lightweight Deep Learning Model for Smartphone," *IJSAT*, 2025.

[35] "SmartAgriDoc: Offline Plant Disease Detection with AI-Powered Mobile App," *IRJET*, vol. 12, no. 5, 2025.

[36] Google Developers Blog, "Learn the steps to build an app that detects crop diseases," *developers.googleblog.com*. [Online]. Available: https://developers.googleblog.com/en/learn-the-steps-to-build-an-app-that-detects-crop-diseases/. Accessed: Nov. 23, 2025.

[37] H. Yu, "A Detailed Introduction to ResNet and Its Implementation in PyTorch," *Medium*. [Online]. Available: https://medium.com/@freshtechyy/a-detailed-introduction-to-resnet-and-its-implementation-in-pytorch-744b13c8074a. Accessed: Nov. 23, 2025.

[38] "Banana Leaf Disease Images," *Mendeley Data*, V1. [Online]. Available: https://data.mendeley.com/datasets/rjykr62kdh/1. Accessed: Nov. 23, 2025.

[39] DigitalOcean, "Writing ResNet from Scratch in PyTorch," *digitalocean.com*. [Online]. Available: https://www.digitalocean.com/community/tutorials/writing-resnet-from-scratch-in-pytorch. Accessed: Nov. 23, 2025.

[40] S. Earman, "Banana Leaf Spot Diseases (BananaLSD) Dataset," *Kaggle*. [Online].

Available: https://www.kaggle.com/datasets/shifatearman/bananalsd. Accessed: Nov. 23, 2025.

[41] S. Kapadnis, "Banana Disease Recognition Dataset," *Kaggle*. [Online]. Available: https://www.kaggle.com/datasets/sujaykapadnis/banana-disease-recognition-dataset. Accessed: Nov. 23, 2025.

[42] M. Bhikadiya, "Plant Disease Detection Using Convolutional Neural Networks with PyTorch," *Analytics Vidhya*. [Online]. Available: https://medium.com/analytics-vidhya/plant-disease-detection-using-convolutional-neural-networks-and-pytorch-87c00c54c88f. Accessed: Nov. 23, 2025.

[43] A. Selek, "plant-disease-classification-pytorch," *GitHub*. [Online]. Available: https://github.com/abdullahselek/plant-disease-classification-pytorch. Accessed: Nov. 23, 2025.

[44] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," *arXiv preprint arXiv:1711.05101*, 2019.

[45] "Deep Learning Based Tomato Leaf Disease Detection and Classification," *IJCRT*, vol. 10, no. 6, 2022.

[46] "Plant Disease Detection using Mask R-CNN," *SSGMCE Shegaon*.

[47] "Interpretable Deep Learning for Banana Leaf Disease Detection Using Dense Convolutional Models," *Cureus Journals*.